

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Nickel rod, 3/8" dia x 5 ft lengths, (200 Grade)

|                                |                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>99188</b>                                                                                         |
| <b>Address</b>                 | ThermoFisher Scientific Australia Pty Ltd<br>5 Caribbean Drive, Scoresby<br>VICTORIA 3179, Australia |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>03 9757 4559 or +613 9757 4559</b>                                            |
| <b>Telephone / Fax Numbers</b> | Tel: 1300 735 292<br>Fax: 1800 067 639                                                               |
| <b>E-mail address</b>          | ANZinfo@thermofisher.com                                                                             |

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

No hazards identified

#### Health hazards

Skin Sensitization  
Carcinogenicity  
Specific target organ toxicity - (repeated exposure)

Category 1  
Category 2  
Category 1

#### Environmental hazards

No hazards identified

### Label Elements



Health Hazard

**Signal Word**

**Danger**

**Hazard Statements**

H372 - Causes damage to organs through prolonged or repeated exposure  
H317 - May cause an allergic skin reaction  
H351 - Suspected of causing cancer

**Precautionary Statements**

P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves  
P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P363 - Wash contaminated clothing before reuse  
P403 - Store in a well-ventilated place  
P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

Toxicity to Soil Dwelling Organisms

## Section 3 - Composition and Information on Ingredients

| Component | CAS No    | Weight % |
|-----------|-----------|----------|
| Nickel    | 7440-02-0 | 100.0    |

## Section 4 - First Aid Measures

|                                            |                                                                                                                                                                                                                         |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                                                                                            |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                                                                                                   |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                                                       |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                         |
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                                                                                                  |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                        |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                    |
| <b>Most important symptoms and effects</b> | May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                  |

## Section 5 - Fire Fighting Measures

**Suitable Extinguishing Media**  
approved class D extinguishers.

**Extinguishing media which must not be used for safety reasons**  
Water may be ineffective.

**Hazardous Decomposition Products**  
Metal fumes and oxides.

**Specific Hazards Arising from the Chemical**  
Thermal decomposition can lead to release of irritating gases and vapors.

**Special protective equipment and precautions for fire fighters**  
As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

**Emergency procedures**  
Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. No special precautions required.

**Environmental Precautions**  
Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up**

**Clean-up methods - small spillage**  
Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. Pick up and transfer to properly labelled containers.

**Clean-up methods - large spillage**  
Typically only supplied in small quantities as packaged goods.  
If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

**Reference to Other Sections**  
Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**  
Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

**Conditions for Safe Storage, Including any Incompatibilities**  
Keep in a dry place. Keep away from acids.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

### Exposure limits

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]  
Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)]

updated in August, 2005. Safe Work Australia **ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace. **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component | Australia                | New Zealand WEL              | ACGIH TLV                  | The United Kingdom                                                            | Germany                                                                                                                               |
|-----------|--------------------------|------------------------------|----------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Nickel    | TWA: 1 mg/m <sup>3</sup> | TWA: 0.005 mg/m <sup>3</sup> | TWA: 1.5 mg/m <sup>3</sup> | STEL: 1.5 mg/m <sup>3</sup> 15 min<br>TWA: 0.5 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 0.03 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 8<br>TWA: 0.006 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 8 |

### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Exposure Controls

#### Engineering Measures

None under normal use conditions.

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

#### Hand Protection

No special protective equipment required

| Glove material    | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|-------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Disposable gloves | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

#### Skin and body protection

Long sleeved clothing

#### Respiratory Protection

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

#### Appearance

Off-white

#### Physical State

Solid

#### Odor

Odorless

#### Odor Threshold

No data available

#### pH

Not applicable

|                                                |                                            |                                          |
|------------------------------------------------|--------------------------------------------|------------------------------------------|
| <b>Melting Point/Range</b>                     | 1455 °C / 2651 °F                          |                                          |
| <b>Softening Point</b>                         | No data available                          |                                          |
| <b>Boiling Point/Range</b>                     | No information available 2730 °C / 4946 °F |                                          |
| <b>Flash Point</b>                             | Not applicable                             | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | Not applicable                             | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available                   |                                          |
| <b>Explosion Limits</b>                        | No data available                          |                                          |
| <b>Vapor Pressure</b>                          | 1 mmHg @ 1810 °C                           |                                          |
| <b>Vapor Density</b>                           | Not applicable                             | Solid                                    |
| <b>Specific Gravity / Density</b>              | 8.90                                       |                                          |
| <b>Bulk Density</b>                            | No data available                          |                                          |
| <b>Water Solubility</b>                        | Insoluble                                  |                                          |
| <b>Solubility in other solvents</b>            | No information available                   |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                            |                                          |
| <b>Autoignition Temperature</b>                | No data available                          |                                          |
| <b>Decomposition Temperature</b>               | No data available                          |                                          |
| <b>Viscosity</b>                               | Not applicable                             | Solid                                    |
| <b>Explosive Properties</b>                    | No information available                   |                                          |
| <b>Oxidizing Properties</b>                    | No information available                   |                                          |
| <b>Other information</b>                       |                                            |                                          |
| <b>Molecular Formula</b>                       | Ni                                         |                                          |
| <b>Molecular Weight</b>                        | 58.69                                      |                                          |

## Section 10 - Stability and Reactivity

|                                         |                                            |
|-----------------------------------------|--------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available |
| <b>Stability</b>                        | Stable under normal conditions.            |
| <b>Conditions to Avoid</b>              | Heat, flames and sparks.                   |
| <b>Incompatible Materials</b>           | None known.                                |
| <b>Hazardous Decomposition Products</b> | Metal fumes and oxides.                    |
| <b>Hazardous Polymerization</b>         | No information available.                  |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

|                            |                                                                  |
|----------------------------|------------------------------------------------------------------|
| <b>(a) acute toxicity;</b> |                                                                  |
| <b>Oral</b>                | Based on available data, the classification criteria are not met |
| <b>Dermal</b>              | No data available                                                |
| <b>Inhalation</b>          | No data available                                                |

| Component | LD50 Oral                 | LD50 Dermal | LC50 Inhalation              |
|-----------|---------------------------|-------------|------------------------------|
| Nickel    | LD50 > 9000 mg/kg ( Rat ) |             | LC50 > 10.2 mg/L ( Rat ) 1 h |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

**(d) respiratory or skin sensitization;**

**Respiratory** No data available  
**Skin** Category 1

**Sensitization** No information available

**(e) germ cell mutagenicity;** No data available

**(f) carcinogenicity;** Category 2

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component | Australia | New Zealand          | New South Wales | Western Australia | IARC     | EU | UK | Germany |
|-----------|-----------|----------------------|-----------------|-------------------|----------|----|----|---------|
| Nickel    |           | Suspected carcinogen |                 |                   | Group 2B |    |    | Cat. 1  |

**(g) reproductive toxicity;** No data available

**(h) STOT-single exposure;** No data available

**(i) STOT-repeated exposure;** Category 1

**Target Organs** No information available.

**(j) aspiration hazard;** Not applicable  
Solid

**Symptoms / effects, both acute and delayed** Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

## Section 12 - Ecological Information

**Ecotoxicity effects**

The product contains following substances which are hazardous for the environment.  
Contains a substance which is: Very toxic to aquatic organisms.

| Component | Freshwater Fish                                                                                                                                     | Water Flea          | Freshwater Algae                            | Microtox |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------|----------|
| Nickel    | LC50: > 100 mg/L, 96h (Brachydanio rerio)<br>LC50: = 1.3 mg/L, 96h semi-static (Cyprinus carpio)<br>LC50: = 10.4 mg/L, 96h static (Cyprinus carpio) | EC50 = 510 µg/L 96h | EC50 = 0.1 mg/L 72h<br>EC50 = 0.18 mg/L 72h |          |

**Persistence and Degradability**

**Persistence**

Insoluble in water.

**Degradability**

Not relevant for inorganic substances.

**Degradation in sewage treatment plant**

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential**

May have some potential to bioaccumulate

**Mobility**

Spillage unlikely to penetrate soil. Is not likely mobile in the environment due its low water solubility

**Endocrine Disruptor Information**

This product does not contain any known or suspected endocrine disruptors

**Persistent Organic Pollutant**

This product does not contain any known or suspected substance

**Ozone Depletion Potential**

This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste from Residues/Unused Products**

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**Other Information**

Chemical wastes should be disposed through a licensed commercial waste collection service. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

## Section 14 - Transport Information

**IMDG/IMO**

Not regulated

**ADG**

Not regulated

**IATA**

Not regulated

**Environmental hazards**

No hazards identified

**Special Precautions**

No special precautions required

**Additional information**

None known

## Section 15 - Regulatory Information

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**National Regulations**

**Australia**

See section 8 for national exposure control parameters.

**Standard for the Uniform Scheduling of Medicines and Poisons**

No poison schedule number allocated.

**Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component          | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|--------------------|-------------------------------------------------------------|------------------------|
| Nickel - 7440-02-0 | Present                                                     | -                      |

**Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

**Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Subject to reporting requirements

| Component          | National pollutant inventory                                                                                                                  |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Nickel - 7440-02-0 | 10 tonne/yr. Threshold category 1<br>2000 tonne/yr. Threshold category 2b<br>60000 MWH. Threshold category 2b<br>20 MW. Threshold category 2b |

## Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

| Component          | Australia | New South Wales | Western Australia | New Zealand          |
|--------------------|-----------|-----------------|-------------------|----------------------|
| Nickel - 7440-02-0 |           |                 |                   | Suspected carcinogen |

## International Inventories

| Component | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|-----------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Nickel    | X    | X     | 231-111-4 | -      | X    | X   | -    | X     | X    |      | X     | KE-25818 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

## Basel convention on the control of transboundary movements of hazardous wastes and their disposal

Not applicable.

| Component | CAS No    | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------|-----------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Nickel    | 7440-02-0 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

## Authorisation/Restrictions according to EU REACH

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Nickel    | -                                                                   | Use restricted. See item 27.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>



## Section 16 - Other Information

### Legend

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>AICS</b> - Australian Inventory of Chemical Substances                                            | <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                                            |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                      | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     |
| <b>IARC</b> - International Agency for Research on Cancer                                            | Predicted No Effect Concentration (PNEC)                                                                                     |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  | <b>ADG</b> Australian Code for the Transport of Dangerous Goods by Road and Rail                                             |
| <b>NZS 5433:2012</b> - Transport of Dangerous Goods on Land                                          | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### **Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

|                         |                  |
|-------------------------|------------------|
| <b>Revision Date</b>    | 20-Nov-2022      |
| <b>Revision Summary</b> | Initial Release. |

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet